

CSE 1201

Selection Sort

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Selection Sort

- Idea:
 - Find the smallest element in the array
 - Exchange it with the element in the first position
 - Find the second smallest element and exchange it with the element in the second position
 - Continue until the array is sorted
- Disadvantage:
 - Running time depends only slightly on the amount of order in the file

Example

8	4	6	9	2	3	1
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Pass 1: search (0 to n-1) lowest and swap with index 0

1	4	6	9	2	3	8
---	---	---	---	---	---	---

Pass 2: search (1 to n-1) lowest and swap with index 1

1	2	6	9	4	3	8
---	---	---	---	---	---	---

Pass 3: search (2 to n-1) lowest and swap with index 1

1	2	3	9	4	6	8
---	---	---	---	---	---	---

Pass 4: search (3 to n-1) lowest and swap with index 1

1	2	3	4	9	6	8
---	---	---	---	---	---	---

Pass 5: search (4 to n-1) lowest and swap with index 1

1	2	3	4	6	9	8
---	---	---	---	---	---	---

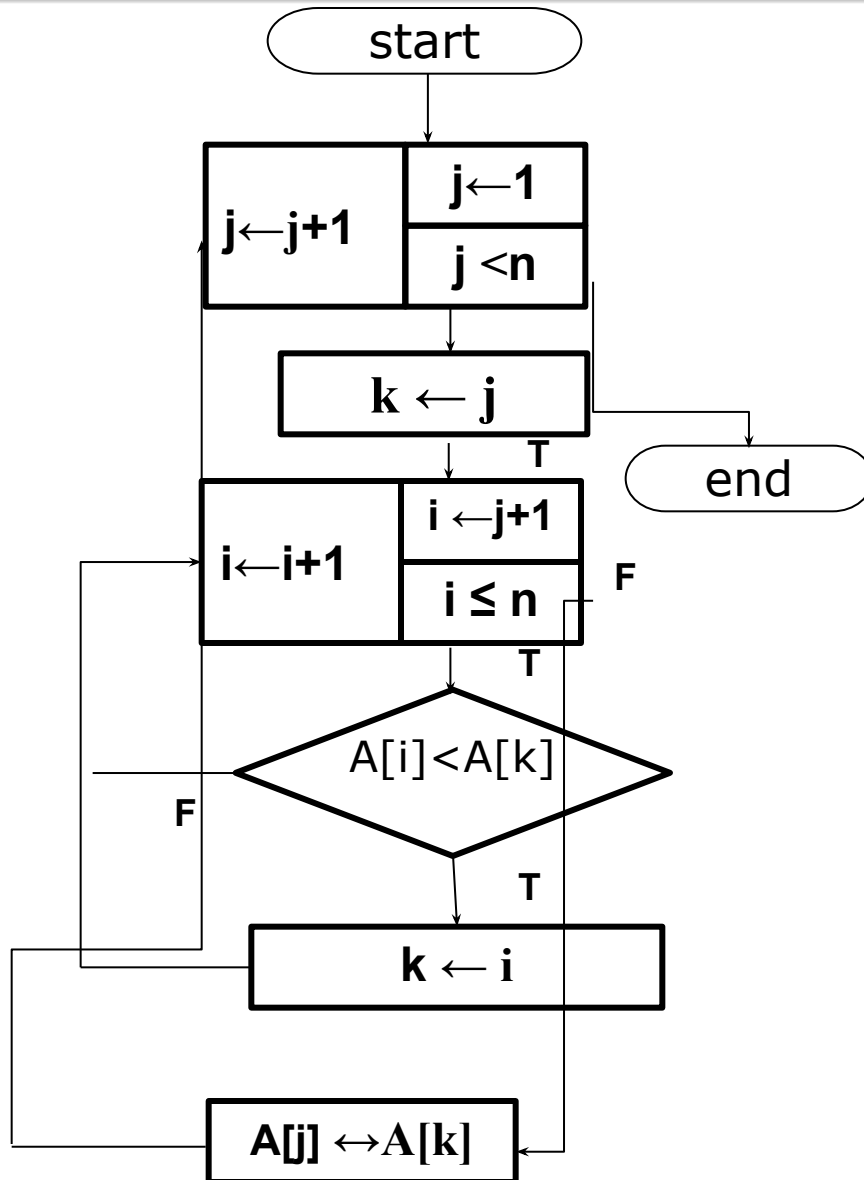
Pass 6: search (5 to n-1) lowest and swap with index 1

1	2	3	4	6	8	9
---	---	---	---	---	---	---

Pass 7: search (6 to n-1) lowest and swap with index 1

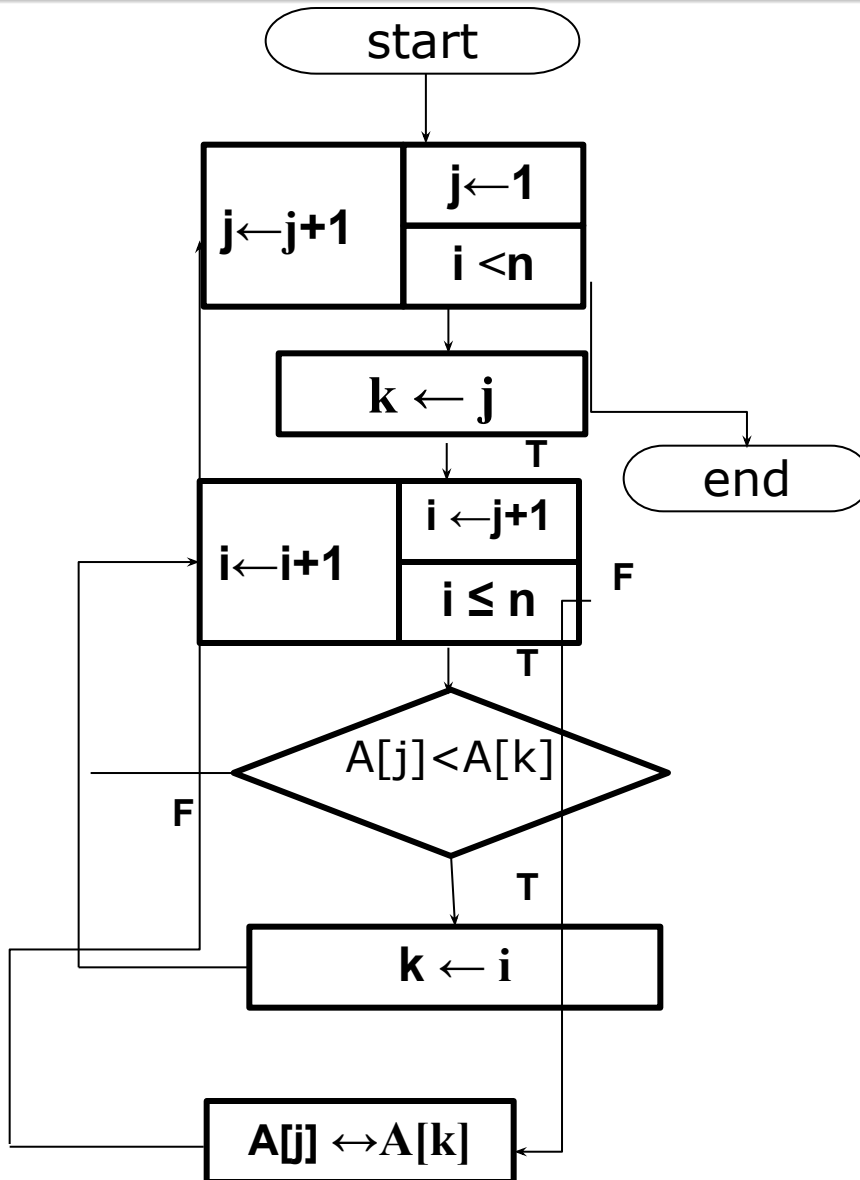
1	2	3	4	6	8	9
---	---	---	---	---	---	---

Selection Sort



8	4	6	9	2	3	1
---	---	---	---	---	---	---

Selection Sort



8	4	6	9	2	3	1
---	---	---	---	---	---	---

```

int main(){
    int i,j,k,t,n;
    int a[7] = {8,4,6,9,2,3,1};
    n=7;
    for(j=0;j<n-1;j++){
        k=j;
        for(i=j+1;i<n;i++){
            if(a[i]>a[k])
                k=i;
        }
        t=a[j];a[j]=a[k];a[k]=t;
    }

    printf("The sorted list\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);
}
  
```

Analysis of Selection Sort

Alg.: SELECTION-SORT(A)

cost times

$n \leftarrow \text{length}[A]$

c_1 1

for $j \leftarrow 1$ **to** $n - 1$

c_2 n

do $\text{smallest} \leftarrow j$

c_3 $n-1$

$\approx n^2/2$
comparisons

for $i \leftarrow j + 1$ **to** n

c_4 $\sum_{j=1}^{n-1} (n - j + 1)$

do if $A[i] < A[\text{smallest}]$

c_5 $\sum_{j=1}^{n-1} (n - j)$

then $\text{smallest} \leftarrow i$

c_6 $\sum_{j=1}^{n-1} (n - j)$

$\approx n$
exchanges

exchange $A[j] \leftrightarrow A[\text{smallest}]$

c_7 $n-1$

$$T(n) = c_1 + c_2 n + c_3 (n-1) + c_4 \sum_{j=1}^{n-1} (n-j+1) + c_5 \sum_{j=1}^{n-1} (n-j) + c_6 \sum_{j=2}^{n-1} (n-j) + c_7 (n-1) = \Theta(n^2) \quad 6$$